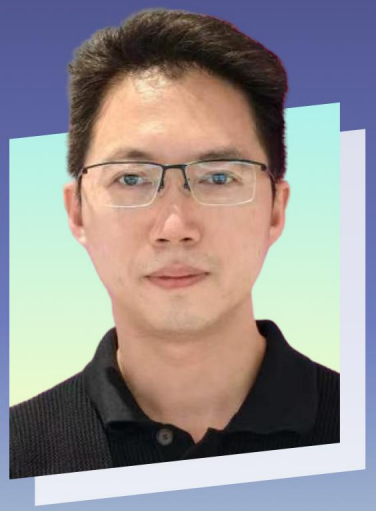
GOSIM®
GLOBAL OPEN-SOURCE INNOVATION MEETUP

GOSIM Shenzhen 2026

Topic

从数据工厂到世界编译器

From the Data Factory to the World Compiler



唐新民 Xinmin Tang

诺亦腾机器人联合创始人

Co-Founder , Noitom Robotics

GOSIM

GOSIM Shenzhen 2026

October 16-17, 2026 Shenzhen, China



GOSIM

GOSIM Shenzhen 2026

From the Data Factory to the World Compiler

Dr. Xinmin Tang

Noitom Robotics, Co-founder



GOSIM 2024: A mocap-based data factory for embodied AI





From motion capture to the frontier of embodied AI

From 13 years of motion capture to the frontier of embodied AI: Building the foundation of intelligent machines.

2012

Noitom was founded

2013



Released the world's first wireless motion capture system

2015



Launched the first motion capture system capable of full-body and finger tracking; gained global attention through Kickstarter

2016-2018



Set a world record for large-scale VR by enabling 100 people to sim-ultaneously en-gage in a shared VR experience. Accumulated deep expertise in operating large-space, multi-user immersive systems.

2019-2022



Introduced hybrid optical-inertial tracking solutions, enabling accurate capture in complex, large-scale environments

2023-2024



Identified a surge in demand from the humanoid robotics and embodied AI sectors; began building motion capture–driven data pipelines to power robot learning

2025

Noitom Robotics spin-off

25000+

Total sales of motion capture products

70%

The choice of 70% of profess-ional mocap users worldwide

50+

Serve customers in over 50 countries

300+

Trusted by over 300 research institutions around the world

500+

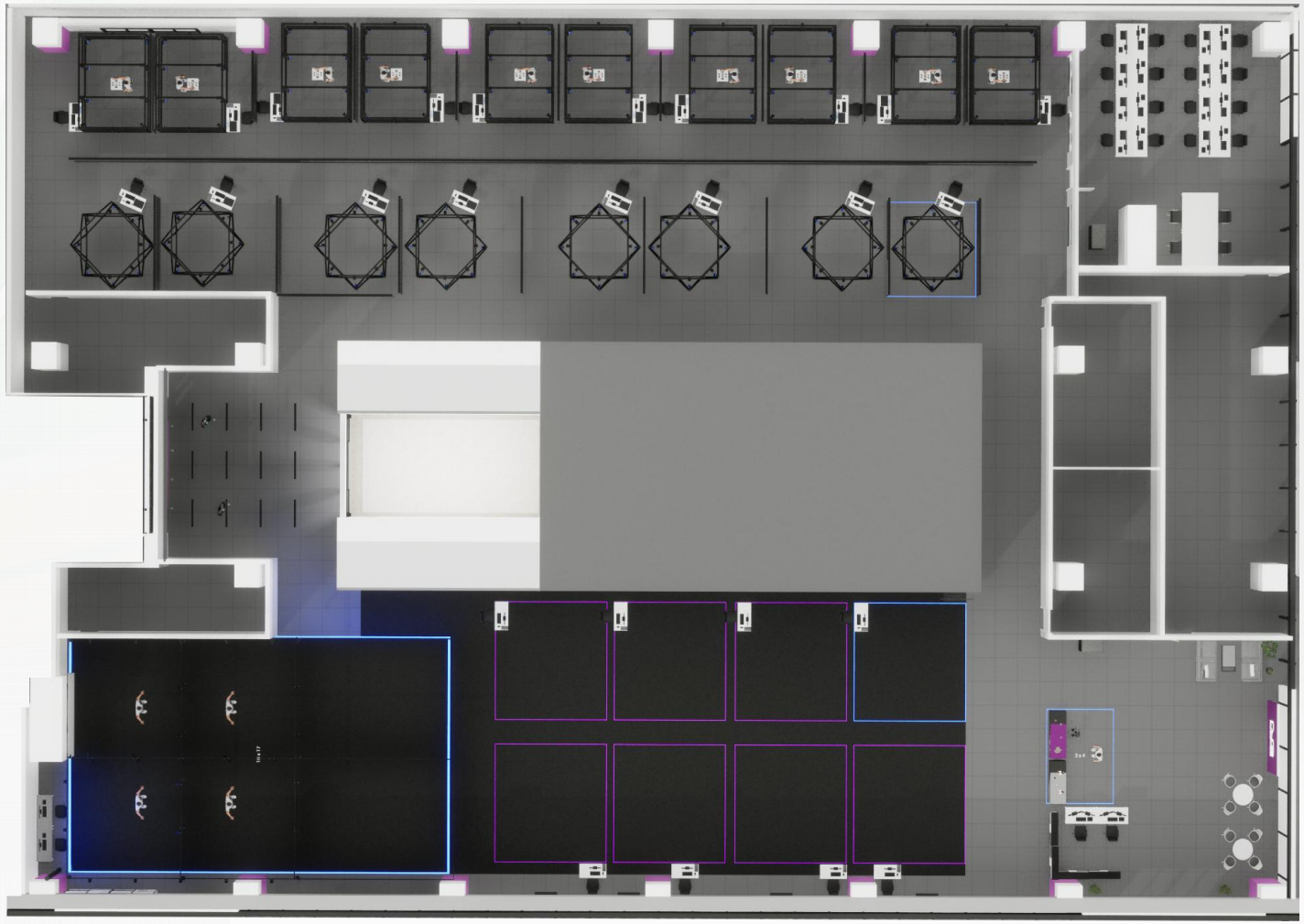
Proven in 500+ commercial projects across sectors

Noitom Robotics data factory overview









Total Space: 5,000 m²+

30+ Standard Single-User Collection Seats (3×4×2.8m)

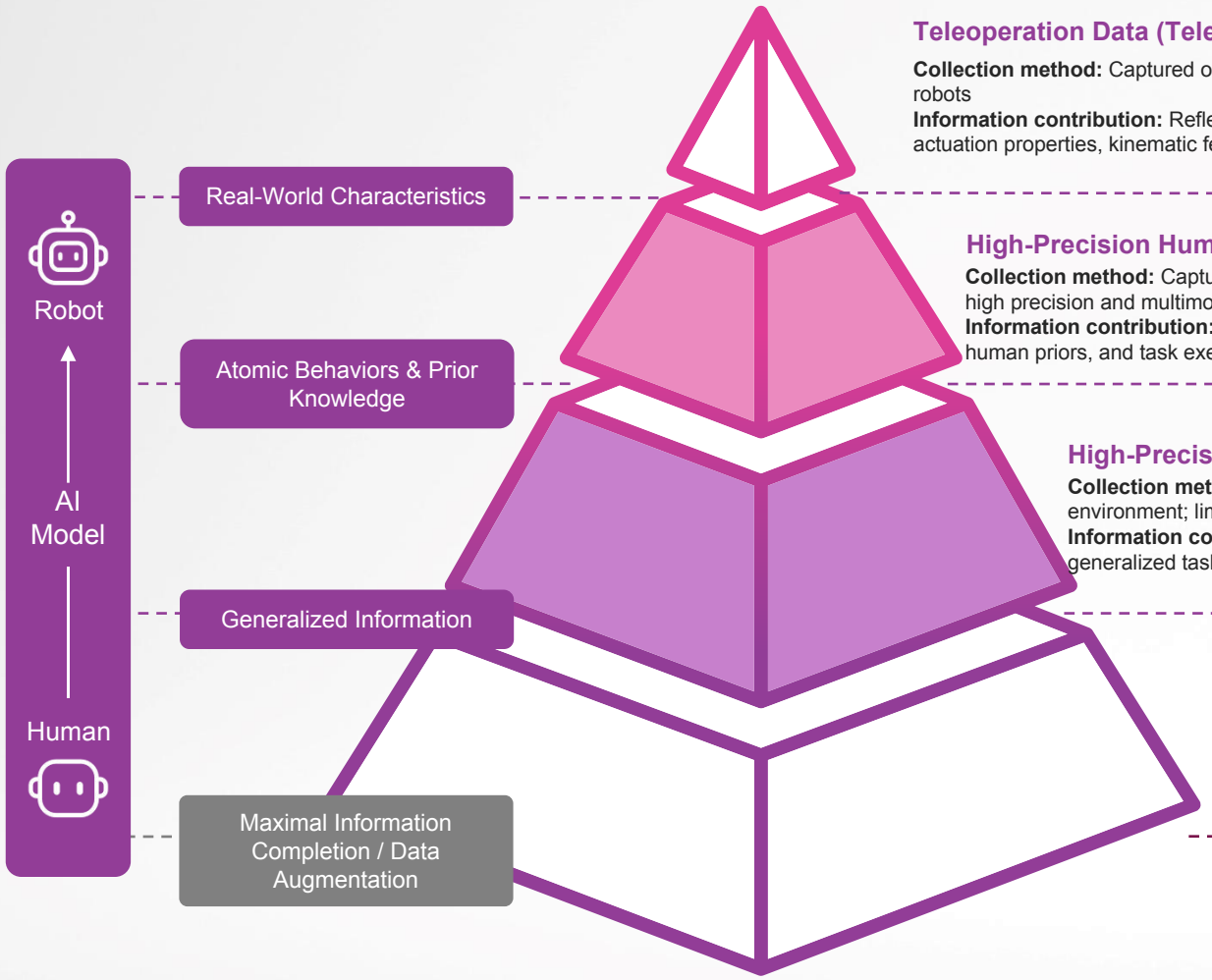
10+ Standard Dual-User Collection Seats (4×4×2.8m)

4 Large-Space Collection Area（100㎡）

Diverse real-world data collection environments spanning over 1,500 m².

Expanding more...

The Embodied AI Data Pyramid



Teleoperation Data (Tele-Op)

Collection method: Captured on the client side through teleoperation of various real robots
Information contribution: Reflects real robot characteristics, true sensor feedback, actuation properties, kinematic features, and mechanical performance

Data output: Over 100 samples per operator per day
Data scale: Hundreds to thousands of hours

High-Precision Human-centric Data (HPHI-OM / MOV)

Collection method: Captured in data factories under controlled environments, with ultra-high precision and multimodal synchronization
Information contribution: Human atomic behaviors (meta motions), scenario-independent human priors, and task execution strategies guided by precise multimodal cues

Data output: 5 hours per operator seat per day
Data scale: Tens of thousands to hundreds of thousands of hours

High-Precision Human-centric Data in the wild (HPHI-ITW)

Collection method: Captured in daily-life or standardized scenarios under un-controlled environment; limited, self-consistent signals derived from real devices and environments
Information contribution: Generalized scenarios, generalized operational goals, generalized task requirements, diverse execution strategies reflecting operator variability

Data output: 5 hours per person per day
Data scale: Millions to tens of millions of hours

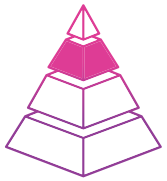
Internet and Synthetic Data

Collection method: Pure software- or document-based collection; includes web-crawled and synthesized data extrapolated from real datasets, and limited RL exploration preserving task features
Information contribution: Broadly supplements human common knowledge, world physical rules, and scales up the dataset to fill gaps in the first three data layers
Data scale: Hundreds of millions to billions of hours

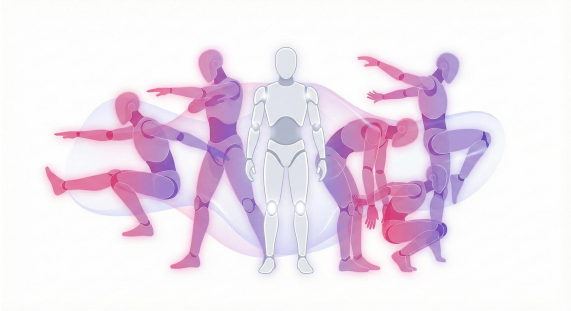
Core Business of Noitom Robotics

Complementary Roles in an Embodied AI Data Stack

HiPHI-MOV | Kinematic Body (The Action Ontology)

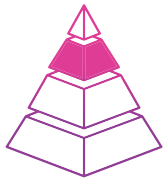


- Delivers massive coverage of the human motion manifold with endless variations and spatial data augmentations.



Strategic Objective: Solves "How the body moves" to establish core motor capabilities.

HiPHI-OM | High-Precision Tasks

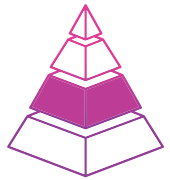


- Focuses on fine-grained manipulation, long-horizon sequences, clear goals, and verifiable state transitions.

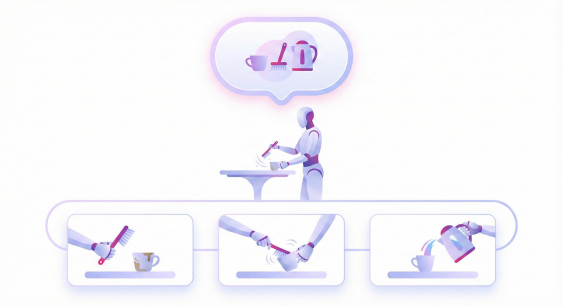


Strategic Objective: Solves "Why actions are taken" to build task structure into the model.

ITW+ | Real-World Experience



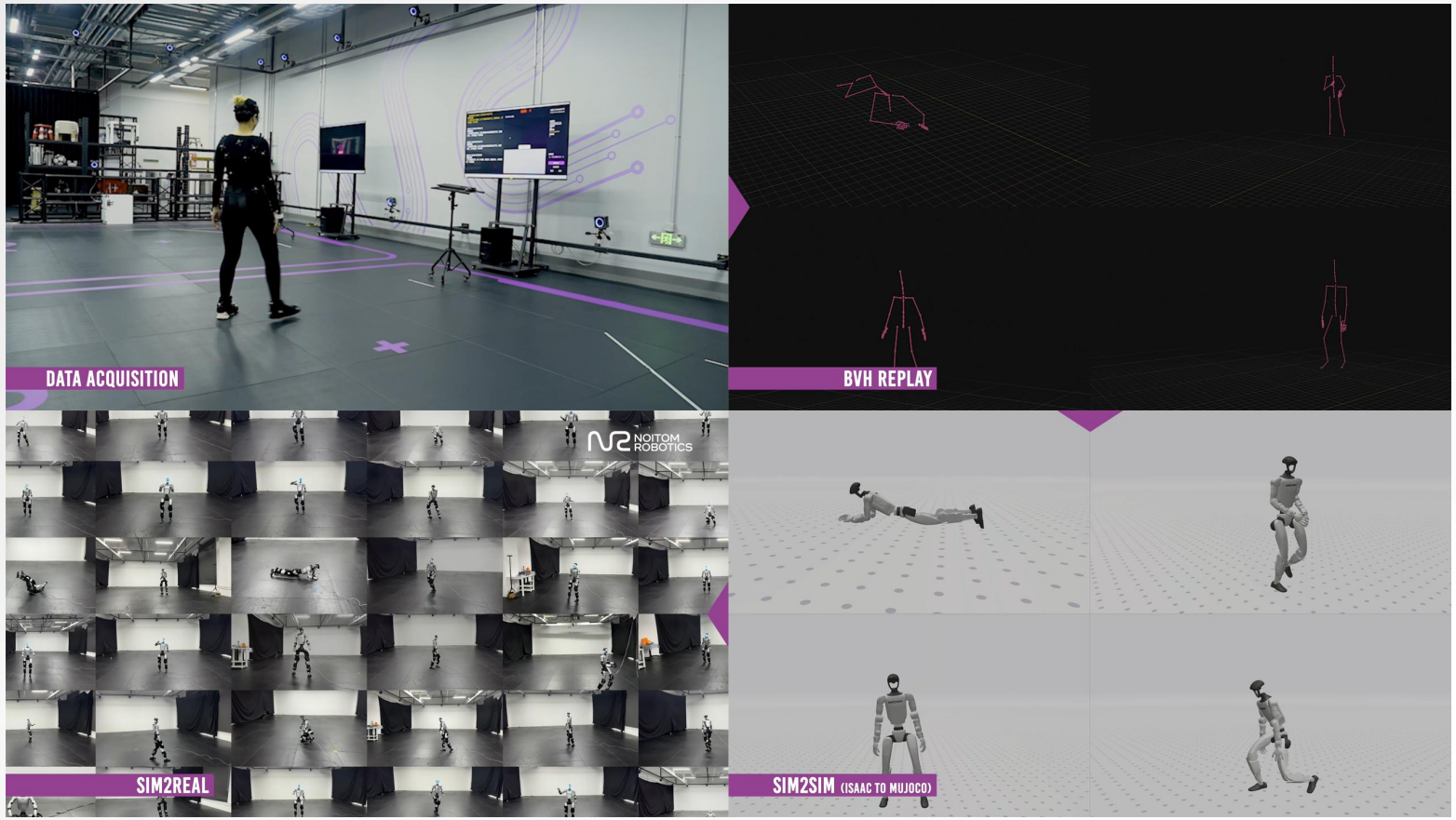
- Captures unconstrained behavior, life-scale scenarios, operator variability, and long-tail distributions.



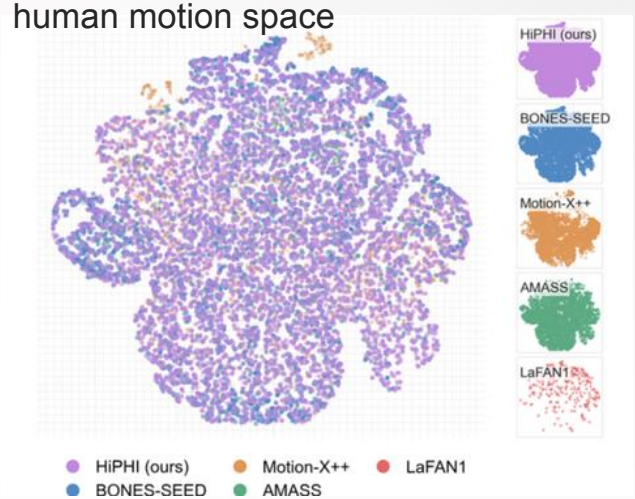
Strategic Objective: Solves "What works in the real world" to guarantee industrial-grade deployment.

HiPHI-MOV (Human-Centric Whole-Body)

A High-Fidelity Whole-Body Motion Dataset Focused on Primitive Human Behaviors



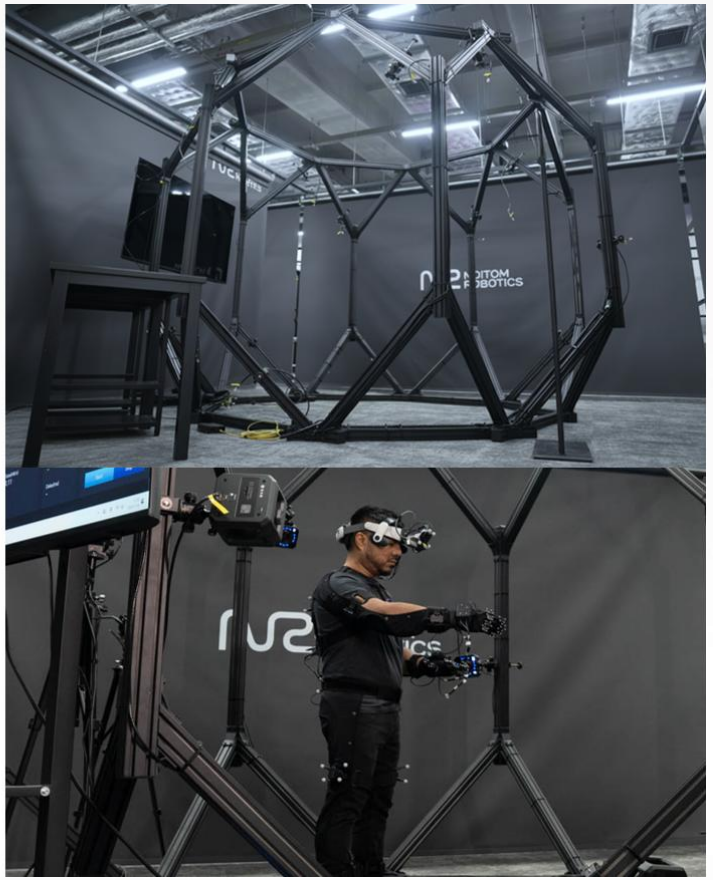
- High-fidelity whole-body motion dataset
- The first industrial-scale dataset of its kind, two orders of magnitude larger than existing datasets.300+ hours publicly released.
- Primitive behaviors covering everything from common motions to long-tail actions, providing the most comprehensive representation of the human motion space



HiPHI-OM (High-Precision Manipulation)

A Fine-Grained Dataset Focused on the Causality of Human Behaviors





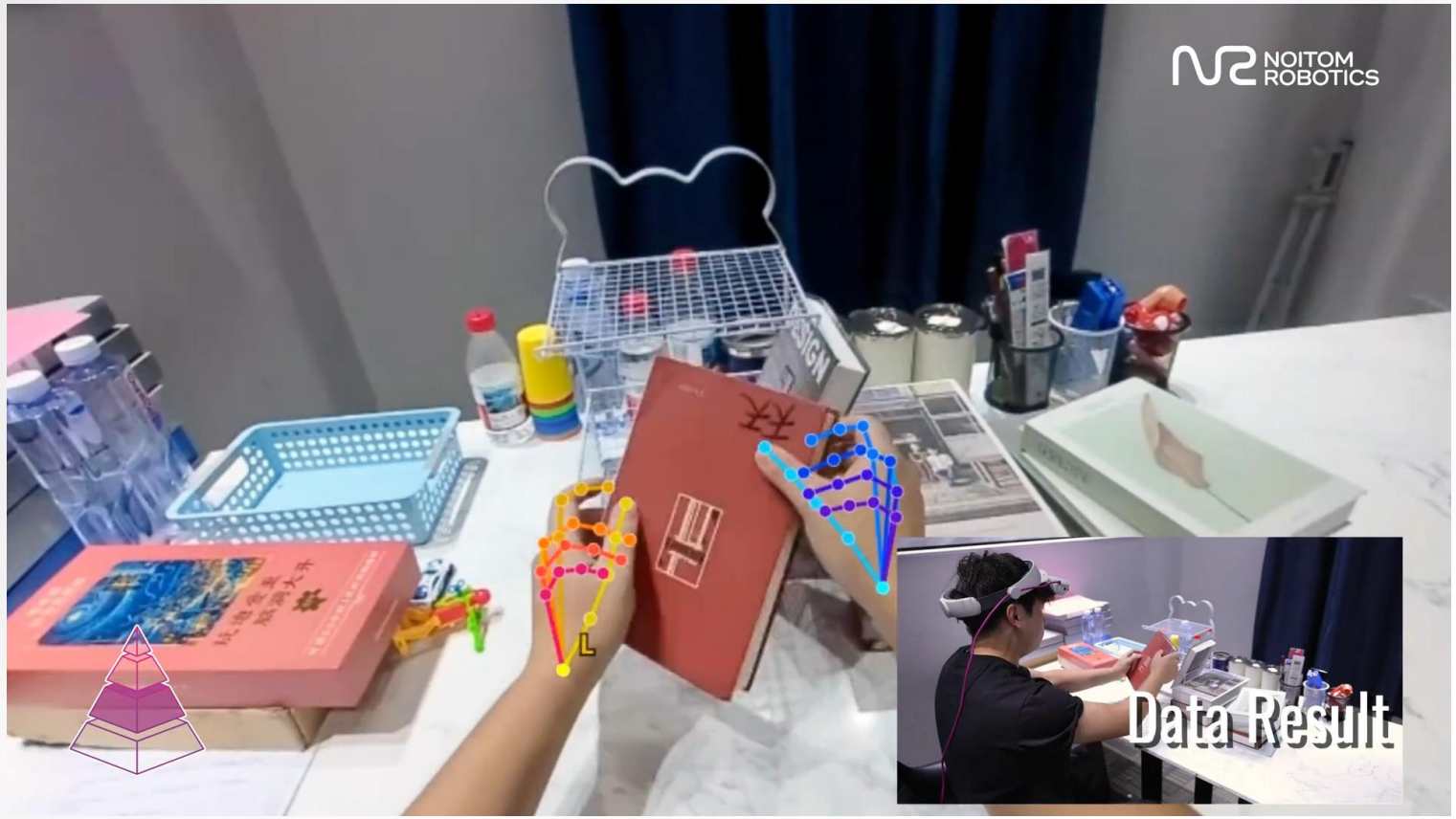
- Ultra-high fidelity, omni-modal
- The world ‘s most comprehensive fully time-synchronized cross-modal dataset. Synchronized modalities include:

First-person RGB-D	Tactile sensing
Third-person RGB	Audio
Hand pose	

- Long-horizon tasks focused on task success
- Designed for multiple downstream applications, including VLA, RL, simulation, and modality completion.

ITW+ (In the Wild)

Human-Centric Data Collection for Visual Generalization in Real-World Environments



- Human-centric data collection in real, uncontrolled environments
- Designed with visual generalization as the primary objective
- Rich multimodal signals

  RGB-D vision

  6DoF wrist pose

- Custom-built hardware, including head-mounted cameras, full-body motion capture, and tactile gloves

Three Priors, One Representation

Factory

Home

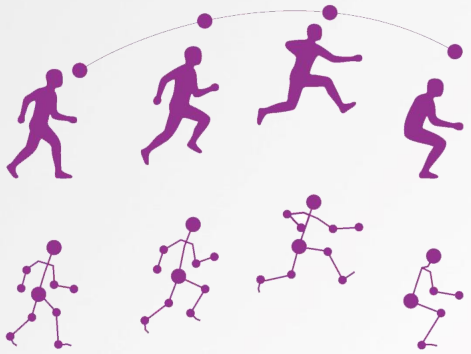
Public Space

Office

ITW

Real-World Distribution Prior

BVH Skeletons



Contact Point

Tracked Object

Tactile

Force

RGB-D



MODALITYNET

A Shared Representation for Physical AI

HIPHI-MOV

Motion Prior

HIPHI-OM

Interaction Prior


modalitynet
Noitom Robotics

ModalityNet as an open research dataset

Noitom Robotics will release 600 hours of high-quality open research data on Hugging Face.

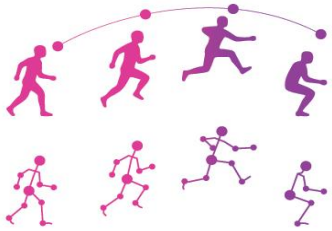
NOITOM ROBOTICS OPENLY RELEASED DATA CONTRIBUTION

600 + Hrs

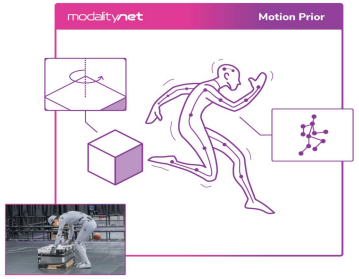
HiPHI is a 617.5-hour optical motion-capture dataset built to make whole-body human motion learnable for humanoid AI. Captured from 102 performers at 90 Hz with sub-millimeter tracking, it includes 245.7 hours of human-object interaction with the object's trajectory and mesh part of the motion record. Coverage is designed around FrameNet-derived motion seeds — the broadest measured coverage among the datasets compared in the paper.



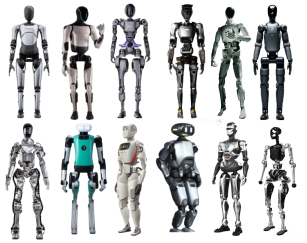
High-Precision whole-body motion dataset, released at a scale exceeding existing classic motion capture datasets such as **AMASS** and **LaFAN**..



By organizing motion into learnable physical behavior representations, it provides a motion foundation for humanoid robots and embodied intelligence.



Comprehensively covering high-precision full-body human motion, hand poses, and the spatial trajectories of interacting objects.



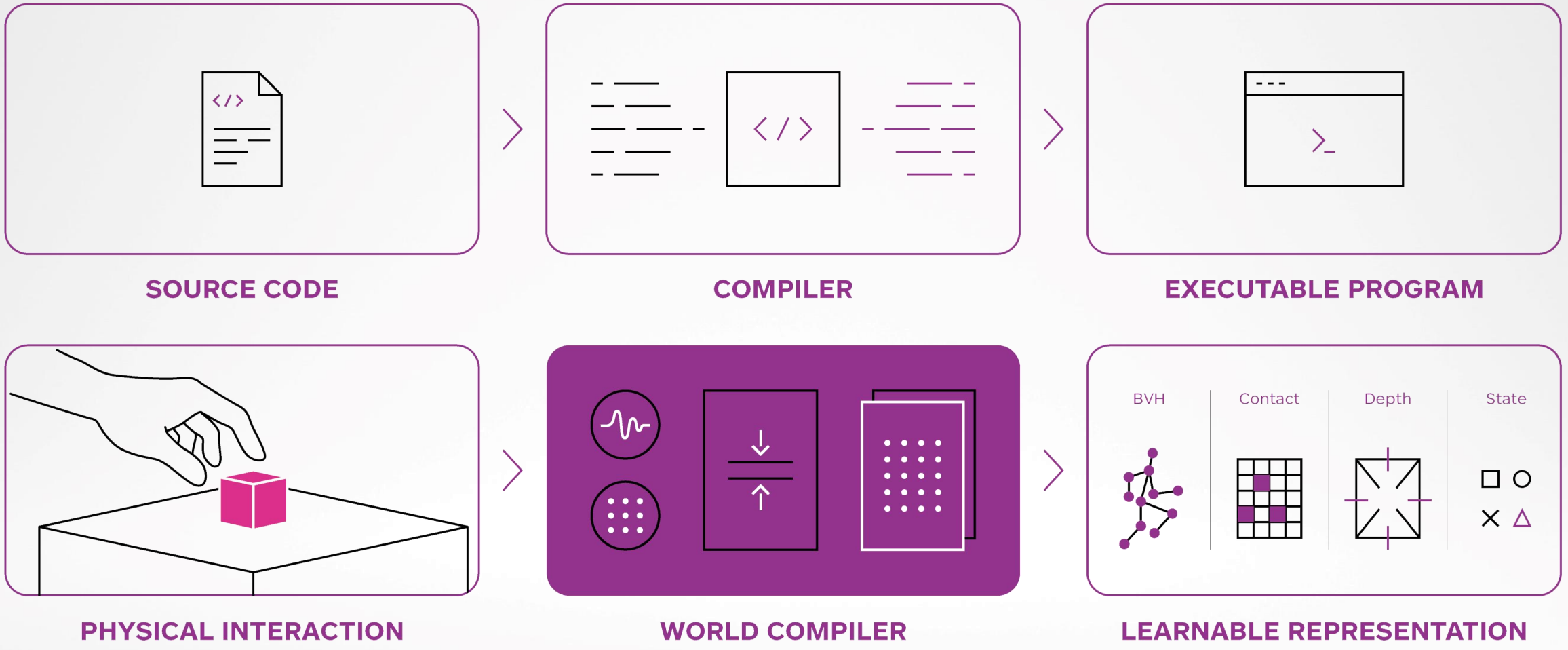
Enables reuse across different robot embodiments.

图中本体仅为示意


00:00:00:00

ModalityNet as the World Compiler

Building a Learnable Physical World



World Compiler: Building a Learnable Physical World

STRUCTURED CORE
HIPHI-MOV & HIPHI-OM

Motion Ground Truth

Walk · Run · Bend/Reach · Squat/Turn

Interaction Ground Truth

Hand-Object · Object State

Multimodal Signals

Visua/Depth · Signals · Tactile

ITW WILD CORPUS
Real-World Raw Data

Home · Office · Factory · Public Space

Validation Loop

1 Mask HIPHI → 2 Reconstruct → 3 Compare

Pred · GT

Learned Compiler

Depth & Hand State · Body Completion · Label Augmentation

COMPILED LEARNABLE OUTPUTS

Depth & Hand State

Body Completion

Action · Object · Target · Relation

Label Augmentation

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Thanks!



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Open Source Robotics

